Cyber-Physical Infrastructure Monitoring: Technical Feasibility, Optimizations, and Edge-Compute Paradigms for Ultra-Low-Cost Event-Driven Bridge Sensor Nodes

Technical Feasibility and Power Budget Analysis

Civil infrastructure monitoring in remote or economically constrained regions requires a departure from traditional, grid-powered, continuous-logging structural health monitoring architectures. The deployment of a distributed, ultra-low-cost, event-driven sensor node network presents a technically viable and fiscally disruptive paradigm for preserving rural bridge networks. To evaluate the physical and operational feasibility of this system, the electrical power budget, wake-up latency limits, and radio frequency compliance standards must be analyzed under realistic field conditions.

The operational profile of the field sensor node relies on an asymmetric power state model where the system spends approximately 99% of its operational lifespan in a microampere deep sleep state. During this quiescent period, the main Xtensa dual-core LX7 CPU of the ESP32-S3 is completely de-powered. Power is conserved by maintaining only the Ultra-Low-Power (ULP) coprocessor, the Real-Time Clock (RTC) peripherals, and the internal 8 KB of RTC fast/slow SRAM. The primary wake-up source is a hardware interrupt routed from the ADXL345 accelerometer to an RTC-capable GPIO pin.

The electrical current consumption is tightly managed across these states to ensure power independence when paired with a small solar energy harvesting system.

Operational Current Budget across Device States

System State	Active Subsystems	Typical Current Consumption	Power Source / Regulation	State Duration
Deep Sleep	RTC Controller, RTC SRAM, ULP, ADXL345 Standby	15\ \mu\text{A} - 40\ \mu\text{A}	LiFePO4 Battery (direct RTC rail)	Variable (up to 99% of lifetime)
Active Sensing	ESP32-S3 Core, ADXL345 Active, HX711, INMP441	20\ \text{mA} - 68\ \text{mA}	LDO Regulator (3.3V active rail)	3.0 - 4.0\ \text{seconds} per vehicle event
Active Telemetry	LoRa SX1276 Transmitter active (+17 dBm to +20 dBm)	110\ \text{mA} - 140\ \text{mA}	Buck-Boost Regulator (high-current rail)	50\ \text{ms} - 120\ \text{ms} (Time-on-Air)
Solar Harvesting	CN3791 MPPT	10\ \text{mA} -	CN3791	Solar peak hours

System State	Active Subsystems	Typical Current Consumption	Power Source / Regulation	State Duration
	Active, 5V 1.5W PV Panel charging	300\ \text{mA} (charging input)	Constant-Current Buck Charger	(diurnal variation)

The system's feasibility depends heavily on the transition latency from deep sleep to active execution. Under a standard cold-boot sequence, the ESP32-S3 exhibits a wake-up latency of approximately 67\ \text{ms} to 96\ \text{ms}. This latency is caused by the hardware reset cycle, ROM bootloader execution, SPI flash memory mapping, and the initialization of the ESP-IDF FreeRTOS scheduler.

During this boot window, the transient physical signals—such as the high-frequency impact acoustic waves or the initial peak acceleration of a vehicle entering the bridge deck—will subside before the sensors can begin active sampling.

To bypass this bottleneck, the ESP32-S3 can run a custom deep sleep wake stub (esp_wake_deep_sleep()) immediately upon exiting sleep. Because this stub is mapped directly into the RTC Fast Memory, it executes within less than 10\ \text{ms} of the hardware interrupt, running before the flash memory is unmapped or the full RTOS is booted. This allows the node to immediately capture raw sensor data during the critical initial moments of a vehicle impact.

Operating within the Indian regulatory framework requires compliance with the Wireless Planning and Coordination (WPC) wing of the Ministry of Communications. Under the gazetted rules of GSR 853(E), the frequency band from 865\ \text{MHz} to 868\ \text{MHz} is de-licensed for short-range, low-power wireless data acquisition devices. The regulatory parameters governing this band dictate strict limits on transmission metrics to prevent local spectrum congestion.

Regulatory Telemetry Standards for GSR 853(E) Compliance

Parameter	Regulatory Constraint (GSR 853(E))	Implemented System Specification	Compliance Verification Mechanism
Frequency Range	865.00\ \text{MHz} - 868.00\ \text{MHz}	865.20\ \text{MHz}, 865.80\ \text{MHz}, 866.40\ \text{MHz}	Channelized LoRa PHY firmware configuration
Max Radiated Power	25\ \text{mW} e.r.p. (14\ \text{dBm}) without APC	+14\ \t[span_79](start_span)[span_79](end_span)ext{dBm} (25\ \text{mW}) transmitter output	Fixed RF output power clamping in SX1276 registers
Adaptive Power Control	Required for transmissions up to 500\ \text{mW}	Not utilized (clamped at standard low-power limit)	Hardware-enforced limit below the 25\ \text{mW} threshold
Occupied Bandwidth	\le 200\ \text{kHz} for tracking/data systems	125\ \t[span_81](start_span)[span_81](end_span)ext{kHz} LoRa Bandwidth	LoRa modulation bandwidth parameter configuration
Duty Cycle Limit	\le 1.0\% for generic, \le 2.5\% for tracking	< 0.1\%[span_82](start_span)[span_82](end_span) under typical traffic	Event-driven transmission throttling algorithm

Parameter	Regulatory Constraint (GSR 853(E))	Implemented System Specification	Compliance Verification Mechanism
		density	

System Vulnerabilities and Hardware Discrepancies

Despite the theoretical benefits of the proposed prototype, several critical vulnerabilities and hardware design discrepancies must be addressed to ensure safe, long-term operation in harsh outdoor environments.

The first major discrepancy is a voltage conflict within the power subsystem:

- **The Battery/Charger Voltage Clash:** The Bill of Materials combines a nominal 3.7\ \text{V} Lithium Iron Phosphate (\text{LiFePO}_4) 18650 cell with a CN3791 MPPT Solar Charger module. This represents a critical chemistry and voltage mismatch. A single-cell \text{LiFePO}_4 chemistry has a nominal voltage of 3.2\ \text{V} and a strict maximum charge termination voltage of 3.6\ \text{V} to 3.65\ \text{V}. However, the CN3791 integrated circuit is designed for standard 3.7\ \text{V} Lithium-Ion/Lithium-Polymer cells, terminating its constant-voltage charging phase at 4.2\ \text{V} \pm 1\%.
- **Overcharging Risk:** Connecting a \text{LiFePO}_4 battery to a 4.2\ \text{V} CN3791 charger will overcharge the cell by 0.6\ \text{V}. This causes rapid electrolyte degradation, outgassing, structural breakdown of the cathode, and potential thermal runaway or combustion under hot outdoor conditions.
- **Load Sharing Limitations:** Low-cost CN3791 charger boards lack dynamic load-sharing (power-path) circuitry. The system load (the ESP32-S3 and its peripherals) is wired directly in parallel with the battery terminals. Consequently, when the solar panel is active, the active current drawn by the microcontroller interferes with the charger's charge-termination logic. The charger may fail to detect when the battery's internal charging current has dropped below the 10\% termination threshold, locking the system in a high-voltage constant-charge state indefinitely and accelerating battery degradation.

Additionally, the selected transducers face physical limitations when deployed for civil infrastructure monitoring.

Standard foil strain gauges, which use a Constantan foil grid on a thin polyimide backing, are highly susceptible to thermal drift. The thermal expansion coefficient of steel or concrete bridge decks differs from that of the Constantan foil.

This mismatch generates a false, temperature-induced strain offset, known as "apparent strain," when the bridge is exposed to ambient temperature changes. Without dynamic temperature compensation, this thermal drift can completely obscure the micro-strain deflections caused by vehicle loads.

Furthermore, standard low-cost adhesives degrade under continuous thermal cycling and moisture exposure, leading to mechanical creep and calibration drift.

The acoustic monitoring subsystem also faces significant limitations:

- **Frequency Band Limitations:** True structural concrete cracking releases stress waves via acoustic emissions (AE) in the high-frequency ultrasonic range, typically from 10\ \text{kHz} to 1,000\ \text{kHz}. The INMP441 digital microphone is designed for the human audible spectrum, with a frequency response of 20\ \text{Hz} to 20\ \text{kHz}. It is physically incapable of capturing high-frequency ultrasonic acoustic emissions from micro-crack formation.
- **Acoustic Noise Pollution:** In outdoor bridge deployments, audible acoustic channels are flooded with environmental noise from wind gusts, heavy rainfall, rushing river water, and

non-structural vehicle impacts. This low-frequency environmental noise masks structural cracking sounds, resulting in a high rate of false-positive triggers unless complex, power-intensive edge filtering is applied.

Value Amplification through Signal Processing and Data Extraction

A key advantage of this edge-computing architecture is its ability to extract multiple high-value structural and traffic metrics from the *existing* hardware sensors, avoiding the cost and complexity of adding external hardware.

Traffic Volumetric and Diurnal Pattern Analysis

Because the ESP32-S3 is triggered by vehicle-induced physical impacts via the ADXL345 interrupt, the node functions as an automated traffic counter. Each successful wake-up event can be recorded as a vehicle passage.
By using the internal RTC to generate a timestamp for each event, the node can log traffic events in the RTC slow memory. This data can be compiled into compact diurnal traffic histograms, compressing a day's traffic flow into a small, 24-byte array that represents hourly passage counts.

```
                TRAFFIC SIGNAL SIGNATURE PROCESSING

[Strain Gauge / HX711] ────────▶ (Zero-Crossing / Hysteresis) ──▶ Axle Count
                            |
                            ▼
[ADXL345 / Accelerometer] ───▶ (Impact Peak-to-Peak Delta)  ──▶ Gross Weight Class
                            |
                            ▼
[INMP441 / Microphone] ────────▶ (Spectral Band Energy Ratio) ──▶ Vehicle Typology
```

Vehicle Speed and Axle Counting Estimation

By analyzing the continuous time-series signal from the foil strain gauge during a vehicle crossing, the node can extract advanced vehicle metrics:

- **Axle Counting:** As a vehicle traverses the bridge, each axle passing over the sensor zone produces a localized deflection peak in the strain gauge output. By applying a zero-crossing or peak-picking algorithm with a hysteresis band to the HX711 output, the node can count these deflection peaks to determine the vehicle's axle count, allowing it to differentiate between two-axle passenger cars and multi-axle commercial trucks.
- **Speed Estimation:** When the physical distance (D) between the accelerometer (mounted near the bridge pier) and the strain gauge (mounted at mid-span) is known, the vehicle's speed (v) can be calculated. By measuring the time-lag (Δt) between the initial ADXL345 motion trigger and the peak strain deflection registered by the HX711, the edge processor can estimate vehicle speed:

$$v = \frac{D}{\Delta t}$$

Vehicle Classification via Acoustic Spectrograms

The INMP441 digital microphone can be used to classify vehicle types based on their acoustic signatures. Heavy commercial trucks, passenger cars, and light two-wheelers generate distinct acoustic spectra due to differences in engine displacement, tire-to-deck friction, and structural rattling.
By running a 256-point Short-Time Fourier Transform (STFT) on the I2S audio stream, the ESP32-S3 can calculate the ratio of low-frequency engine rumble (20\ \text{Hz} - 250\ \text{Hz}) to high-frequency tire noise (1\ \text{kHz} - 8\ \text{kHz}). This spectral energy ratio allows the system to classify vehicles into distinct weight and size categories without needing a complex neural network.

Dynamic Modulus of Elasticity Correction

By correlating the ambient temperature (read from the ADXL345's internal temperature register or an external sensor) with the bridge's baseline resonant frequency (f_n), the system can correct for seasonal material changes. The Young's modulus of concrete (E) varies with temperature, causing the bridge's natural resonant frequency to drift over seasons even in the absence of structural damage.
Using the existing temperature data to apply a regression correction to f_n ensures that alerts are triggered only by actual structural degradation, preventing false alarms caused by seasonal thermal variations.

Expanded Low-Budget Sensor Integration

To increase the structural health monitoring capabilities of the platform while staying within a tight budget, several strategic hardware upgrades can be integrated into the sensor node.

Hydrological and Bridge Scour Sensing

Adding hydrological and bridge scour monitoring provides high value for bridges spanning active waterways. Heavy rain and flooding accelerate riverbed erosion around bridge bents and piers, a process known as hydraulic scour, which is a leading cause of bridge failures.
By mounting a waterproof ultrasonic sensor underneath the bridge deck and pointing it downward, the node can measure water clearance heights during normal flow, and monitor scour depths during high-flow flood events.

```
               BRIDGE SCOUR AND HYDROLOGICAL SENSOR

 [span_45](start_span)[span_45](end_span)          [BRIDGE DECK]
               ┌──────────────────────────┐
               │ Edge Sensor Node │
               └────────[span_46](start_span)[span_46](end_span)───┬──────────┘
                   │ (Waterproof Sonar)
                   ▼
                 [WATER LEVEL]
                 ~ ~ ~ ~ ~ ~ ~ ~
```

```
              /[span_47](start_span)[span_47](end_span)            \
             /   [PIER]     \
            /      |         \
           /       |          \
          /        |           \
         /    ┌────────┴────────┐        \
[RIVERBED] ───────/_ _ | Scour Hole|  _ _ _ _\─────── [ORIGINAL RIVERBED]
              └─────────────────┘
```

The waterproof ultrasonic sensor must be carefully selected to balance range, blind spot constraints, and power efficiency:

- **JSN-SR04T Sensor:** This split-transducer sensor separates its electronics from an IP67-rated transducer on a 2.5-meter cable. It has an effective range of 23\ \text{cm} to 600\ \text{cm}. However, it suffers from a large 20\ \text{cm} to 23\ \text{cm} blind zone, meaning it cannot accurately measure water levels that rise close to the sensor. It also has a wide 75^\circ beam angle, which can cause false echoes from nearby bridge piers or abutments.
- **DFRobot A02YYUW Sensor:** This sensor integrates a single IP67-sealed transducer and uses internal filtering to reduce the blind spot to just 3\ \text{cm}. It provides an effective range of 3\ \text{cm}[span_126](start_span)[span_126](end_span) to 450\ \text{cm} and outputs distance data over a low-power UART interface. This UART interface reduces CPU processing overhead compared to the GPIO pulse timing required by the JSN-SR04T.

By pairing these sonar measurements with the accelerometer's resonant frequency tracking, the system can cross-validate structural integrity: as scour depth increases and destabilizes a pier's foundation, the bridge's fundamental resonant frequency shifts downward, signaling structural distress.

High-Precision Thermal Drift Compensation

To correct for temperature-induced strain errors, a second "dummy" foil strain gauge can be integrated onto an unstressed piece of identical bridge material within the sensor housing. Wiring the active strain gauge and the dummy strain gauge into a half-bridge Wheatstone configuration cancels out the temperature-induced expansion, as both gauges experience identical thermal drift.
This ensures that the HX711 differential ADC measures only the mechanical strain caused by traffic loads.

Strategic Low-Cost Hardware Upgrades

Upgrade Component	Part Number / Model	Key Structural Contribution	Unit Cost (INR)	Sleep Power Impact
Waterproof Sonar	DFRobot A02YYUW	Real-time scour depth and water level monitoring	₹800	Nil (powered down in deep sleep via external MOSFET)
Dummy Strain Gauge	SMBF-SG-120-10	Half-bridge thermal drift	₹124	Nil (passive resistive bridge

Upgrade Component	Part Number / Model	Key Structural Contribution	Unit Cost (INR)	Sleep Power Impact
		cancelation		component)
Waterproof Thermistor	DS18B20	Concrete dynamic elasticity correction	₹110	1\ \mu\text{A} (standby leakage)
Power-Path Switch	DMG2302 (N-Ch MOSFET)	Power gates active sensors during sleep	₹15	Nil (isolation switch)
LiFePO4 Charger	TP5000 / SD30CRMA	Safe 3.6\ \text{V} solar battery charging	₹180	Nil (charging input path only)

Comparative Analysis of Prior Systems and Intellectual Property

To ensure a unique patent position and build upon validated civil engineering practices, the proposed system is compared against existing state-level projects and patent holdings.

Academic and State-Level Research Projects

- **The Georgia DOT / Kennesaw State University Initiative:** Deployed on scour-critical bridges in Georgia, this system utilized JSN-SR04T ultrasonic sonar sensors and ADXL345 accelerometers managed by an ESP32. The node used ESP-NOW for local sensor-to-gateway links and a cellular SIM7000 module for cloud uploading. This project confirmed the inverse relationship between foundation scour and the bridge's resonant frequency, validating the diagnostic power of low-cost edge accelerometers. However, the system relied on cellular uplinks and did not utilize low-latency deep sleep wake stubs, resulting in higher power requirements.
- **The California Caltrans BRACE2 Platform:** Developed by UC Berkeley, the Bridge Rapid Assessment Center for Extreme Events (BRACE2) utilizes event-triggered sensors to enable rapid post-earthquake inspections across 22 bridges. Moving from periodic, time-based schedules to condition-based inspections reduced unnecessary bridge closures and optimized emergency resource allocation. However, the platform relies on high-resolution seismic instrumentation, making it too expensive for rural deployments.
- **The European Horizon Europe FUTURAL Project:** This project developed a crowdsensing framework for rural bridges, using USB-C accelerometers connected to transit vehicles to measure bridge vibrations indirectly. Autoencoder neural networks process the vibration data, identifying structural anomalies using reconstruction error. While highly scalable, the indirect sensing model cannot track localized structural changes over time as accurately as fixed, direct-contact sensors.

Intellectual Property and Patent Landscape

- **CN108279294B (Steel Bridge Non-Destructive Monitoring System):** This patent describes a non-destructive monitoring system for steel bridges using a high-performance TMS320F28335 DSP controller. It manages a solar charging unit, a multi-channel

vibration sensor, a pulse laser generator, a capacitive air-coupled sensor, and GPS positioning units to monitor crack propagation and rotational displacement. While comprehensive, the system's reliance on power-intensive laser and air-coupled sensors makes it unsuitable for low-power, solar-harvesting edge nodes.
- **US7596470B2 / WO2005031501A2 (Diagnostic Patch Network):** This patent describes a structural health monitoring network comprising a plurality of diagnostic patches attached to a host structure. Each patch operates as both a transmitter and a receiver, utilizing integrated piezoelectric devices to actuate and capture Lamb waves. This allows the system to identify internal crack propagation using high-frequency ultrasonic waveforms.

The proposed event-driven platform can carve out a unique patent position by focusing on its hardware-in-the-loop wake-up pipeline: using a passive MEMS accelerometer interrupt to trigger an ESP32-S3 wake stub, which performs on-chip frequency-domain and rainflow damage calculations before selectively transmitting data via LoRa.

Recent Research Ideas and Advanced Implementations

The system's efficiency can be enhanced by incorporating recent advances in embedded systems, such as TDMA-based LoRa scheduling, low-latency wake stubs, and on-chip Rainflow cycle counting.

TDMA-Based LoRa Scheduling to Prevent Packet Collisions

When multiple sensor nodes are deployed along a single bridge or across a cluster of nearby structures, uncoordinated event-driven transmissions can cause packet collisions at the gateway, leading to data loss. To address this, the communication layer can implement a Time-Division Multiple Access (TDMA) scheduling protocol over the LoRa physical layer.

```
                      TDMA LORA FRAME ARCHITECTURE

┌──────────────────────────────────────────────────────────────────────
──────────────────┐
│                        TDMA Frame Duration                          │
└──────────────────────────────────────────────────────────────────────
──────────────────┘
│  Beacon  │  Slot 0  │  Slot 1  │  Slot 2  │ Guard │ Silence/Sleep│
│ (Gateway)│  (Node 1)  │  (Node 2)  │  (Node 3)  │ Interval │  (Windows) │
└───────────┴──────────────┴──────────────┴──────────────┴──────────┴──
──────────────────┘
```

At the start of an operational frame, the gateway broadcasts a synchronization beacon. Each node uses this beacon to align its internal clock and calculate its assigned slot start time (t_{slot}) using a configured index (i):

$$t_{slot} = t_{epoch} + i \cdot T_{slot}$$

Where t_{epoch} is the beacon reception timestamp, and T_{slot} is the calculated slot duration.

To prevent collisions, the slot duration is calculated using the packet's physical Time-on-Air (T_{ToA}), transceiver state transition delays (T_{RF}), and a guard interval (T_{guard}) to absorb clock drift:

$$T_{slot} = T_{ToA} + T_{RF} + T_{guard}$$

The LoRa physical layer Time-on-Air is calculated using the configured preamble length (N_{pre}) and symbol duration (T_s):

$$T_{pre} = (N_{pre} + 4.25) \cdot T_s$$
$$N_{pl} = 8 + \max\left(\left\lceil \frac{8PL - 4SF + 28 + 16CRC - 20IH}{4(SF - 2DE)} \right\rceil \cdot (CR + 4), 0\right)$$
$$T_{pl} = N_{pl} \cdot T_s$$
$$T_{ToA} = T_{pre} + T_{pl}$$

Where PL is the payload length in bytes, SF is the Spreading Factor, CR is the Coding Rate, CRC is the cyclic redundancy check flag, IH is the implicit header mode flag, and DE is the low data rate optimization flag. By utilizing FreeRTOS software timers to enforce these transmission slots, the system eliminates packet collisions, achieving a packet delivery ratio (PDR) exceeding 98\%.

Low-Latency ESP32-S3 Deep Sleep Wake Stub

To capture structural vibration and deflection data immediately upon vehicle impact, the ESP32-S3 must execute its sensor initialization within the custom deep sleep wake stub (esp_wake_deep_sleep()). This code runs directly from the RTC Fast Memory while the main CPU registers and SPI flash remain in their reset, unmapped states.

Implementing a wake stub on the ESP32-S3 requires adherence to several strict architectural constraints:

- **Memory Isolation:** The stub code must be declared with the RTC_IRAM_ATTR attribute. Any persistent variables or read-only string data must be explicitly marked with RTC_DATA_ATTR or RTC_RODATA_ATTR, forcing their allocation into the 8 KB RTC memory. Accessing standard SRAM or flash memory inside the stub will cause a processor panic.
- **No Library Calls:** Standard ESP-IDF or Arduino API functions—such as delay() or analogRead()—cannot be called inside the stub. Only ROM-implemented functions (e.g., ets_printf, ets_delay_us) are accessible.
- **Hardware Initialization:** All peripherals must be initialized manually using direct register writes via the low-level register interface.
- **Linker File Naming:** To simplify memory allocation, source files can be prefixed with rtc_wake_stub (e.g., rtc_wake_stub_sensor.c), prompting the linker to automatically compile their entire contents into the RTC memory.
- **Kconfig Dependencies:** If the Kconfig option CONFIG_ESP32S3_RTCDATA_IN_FAST_MEM is enabled, data marked with RTC_DATA_ATT[span_56](start_span)[span_56](end_span)[span_63](start_span)[span_63](end_span)R is moved to the RTC Fast Memory segment. Because this segment is only accessible by the primary core (PRO_CPU), this option requires enabling CONFIG_FREERTOS_UNICORE.

The C-based implementation below demonstrates how to configure a custom wake stub on the ESP32-S3. This stub initializes the hardware, performs a rapid analog threshold check, and determines whether to boot the full operating system or return to deep sleep.

```
#includ[span_171](start_span)[span_171](end_span)[span_173](start_span)[span_173](end_span)[span_175](start_span)[span_175](end_span)e <stdint.h>
#include "esp_sleep.h"
```

```
#include "esp_attr.h"
#include "rom/ets_sys.h"
#include "hal/gpio_ll.h"
#include "soc/rtc_cntl_reg.h"
#include "soc/rtc_io_reg.h"

// Hardware parameters for the sensor node
#define EXT_WAKEUP_GPIO_PIN 12
#define
ACCEL_PEAK_THRESHO[span_58](start_span)[span_58](end_span)[span_65](start_span)[sp
an_65](end_span)LD 1200

// Persistent counters placed in RTC slow memory
static RTC_DATA_ATTR uint32_t wake_cycle_count = 0;
static RTC_DATA_ATTR uint32_t validated_impact_count = 0;

// Read-only log strings stored in RTC memory
static RTC_RODATA_ATTR const char fmt_wake_log[] = "[RTC STUB] Wake event detected.
Total Cycles: %d\n";
static RTC_RODATA_ATTR const char fmt_valid_log[] = "[RTC STUB] Signal validated
(%[span_38](start_span)[span_38](end_span)d). Triggering full OS boot.\n";
static RTC_RODATA_ATTR const char fmt_reject_log[] = "[RTC STUB] Signal below threshold.
Returnin[span_39](start_span)[span_39](end_span)g to deep sleep.\n";

// Custom wake-up stub function declaration
void RTC_IRAM_ATTR esp_wake_deep_sleep(void);

void RTC_IRAM_ATTR esp_wake_deep_sleep(void) {
   // 1. Run the default system wake-up initialization
   esp_default_wake_deep_sleep();

   // 2. Increment the persistent wake-up counter
   wake_cycle_count++;
   ets_printf(fmt_wake_log, wake_cycle_count);

   // 3. Read the state of the wake-up pin directly from the RTC register
   // This bypasses the unmapped digital GPIO driver
   uint32_t pin_state = (REG_GET_FIELD(RTC_GPIO_IN_REG, RTC_GPIO_IN_NEXT) >>
EXT_WAKEUP_GPIO_PIN) & 0x01;

   if (pin_state == 1) {
      // High-energy event verified. Print log and return to allow full boot
      validated_impact_count++;
      ets_printf(fmt_valid_log, validated_impact_count);
      return;
   } else {
      // Transient noise detected. Print log and prepare to return to sleep
      ets_printf(fmt_reject_log);
```

```
        // Feed the watchdog timer to prevent system resets during register writes
        REG_WRITE(TIMG_WDTFEED_REG(0), 1);

        // Wait for the UART TX buffer to empty before shutting down
        while (REG_GET_FIELD(UART_STATUS_REG(0), UART_ST_UTX_OUT)) {
            ets_delay_us(100);
        }

        // Re-align the RTC control entry address back to this wake-up stub
        REG_WRITE(RTC_ENTRY_ADDR_REG, (uint32_t)&esp_wake_deep_sleep);

        // Clear power state register masks and trigger deep sleep
        CLEAR_PERI_REG_MASK(RTC_CNTL_STATE0_REG, RTC_CNTL_SLEEP_EN);
        SET_PERI_REG_MASK(RTC_CNTL_STATE0_REG, RTC_CNTL_SLEEP_EN);

        // Lock execution in an infinite loop until the CPU enters sleep
        while (1) {
            ;
        }
    }
}
```

On-Chip Rainflow Cycle Counting and Fatigue Estimation

Rather than transmitting raw, high-frequency strain waveforms over bandwidth-limited LoRa channels, the edge-computing node can run a local Rainflow counting algorithm. This algorithm processes raw strain data from the foil strain gauges, identifying closed stress-strain hysteresis loops to compile stress range histograms.

The algorithm processes a continuous sequence of turning points (local peaks and troughs) and applies the standard ASTM E 1049 four-point counting method. Given four consecutive turning points (A, B, C, and D), a closed inner cycle is identified between B and C if the following mathematical condition is satisfied:

$$\min(B,C) \ge \min(A,D) \quad \land \quad \max(B,C) \le \max(A,D)$$

```
                FOUR-POINT RAINFLOW COUNTING DECISION
          Turning Points: A, B, C, D in continuous time-history


                           * B
                          / \
                         /   \     * D
                     A  *     \   /
                               \ /
                                * C


          Is Cycle B-C closed?
          Condition: min(B,C) >= min(A,D) AND max(B,C) <= max(A,D)
```

```
│  YES: Record cycle B-C and    │
│  prune points from the buffer │
└─────────────────────────────────────┘
```

Once a cycle is identified, the stress range ($\Delta \sigma = \vert{}B - C\vert{}$) is calculated, logged in a compact on-chip histogram, and points B and C are pruned from the turning point buffer using RFC_tp_prune() logic to prevent memory overflows.
By accumulating these cycles over time, the node calculates cumulative structural fatigue damage (D) using Palmgren-Miner's linear damage rule:

$$D = \sum_{i=1}^{k} \frac{n_i}{N_i}$$

Where n_i is the number of cycles accumulated at stress range $\Delta \sigma_i$, and N_i is the number of cycles to failure at that same stress range, derived from the material's S-N curve (Wöhler curve). This edge-calculated fatigue damage metric reduces the data transmission payload from several megabytes of raw waveform data to a single 4-byte floating-point number.

Structural Recommendations and Operational Roadmap

To transition the proposed prototype from a lab setup to a reliable, long-term field deployment, several design updates are recommended:

1. **Power Subsystem Redesign:**
   - Replace the single-chemistry CN3791 charging module with a dedicated LiFePO_4 charge controller, such as the TP5000 or SD30CRMA, to limit the maximum charging voltage to a safe $3.6\ \text{V}$.
   - Integrate an active, low-loss MOSFET-based load-sharing switch (power-path management) to isolate the ESP32-S3's active current draw from the charger's termination logic, ensuring clean termination cycles.
2. **Transducer Optimization:**
   - Replace the single foil strain gauge with a half-bridge Wheatstone configuration utilizing an active gauge and an identical, unstressed dummy gauge to compensate for thermal expansion.
   - Mount both gauges to the structural steel or concrete deck using a high-temperature, moisture-resistant epoxy adhesive (such as EP310N) to prevent mechanical creep and bonding degradation under continuous thermal cycling.
3. **Expanded Environmental Monitoring:**
   - Integrate a waterproof ultrasonic sensor—such as the DFRobot A02YYUW—to monitor river clearance levels and pier scour depths during flood events, establishing a physical correlation between scour depth and natural frequency shifts.
4. **Software and Edge Compute Optimization:**
   - Implement a low-latency ESP32-S3 deep sleep wake stub in RTC Fast Memory to validate trigger signals in less than $10\ \text{ms}$, filtering out false alarms before booting the main system.
   - Execute on-chip rainflow counting algorithms to compile stress range histograms locally, reducing the transmitted payload to a single, cumulative structural damage index.

Implementing this operational roadmap will yield a robust, WPC-compliant, and self-sufficient cyber-physical monitoring platform, providing continuous, cost-effective structural safety tracking for vulnerable rural infrastructure assets.

Works cited

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